Ashna Islam

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3rd year B.Tech CSE (Cybersecurity) | IEM Kolkata · CGPA 9.10 | IEEE CS Chapter Chair

EDUCATION

B.Tech – Computer Science & Engineering (Cybersecurity) | Institute of Engineering and Management, Kolkata (School of University of Engineering and Management, Kolkata) | (Jun 2024 – Pursuing)

- **CGPA 9.10** (upto 4th Semester) | **2nd rank in department (2025-2026)**

Shri Shikshayatan School, Kolkata | Class XII: 80.1% | Class X: 93.2%

INTERNSHIPS

Cybersecurity Intern | West Bengal Electronics Industry Development Corporation Limited | (June 2026 - July 2026)

- Administered Linux systems using file permissions, package management, and SELinux.
- Analyzed system logs, AWStats reports, and database security.
- Automated administrative tasks using Bash scripting.

Research Project Intern | IEDC Lab, IEM Kolkata , Dept. of CST / CSIT / CSE (CS & NW) | (Dec 2025 – Jan 2026)

- Developed a Neural Network-based crop recommendation system using soil, temperature, humidity, rainfall, and NPK parameters. Performed data preprocessing and feature engineering, deployed the ML backend on AWS, and implemented an IoT edge server using Raspberry Pi for real-time agricultural decision support.

STUDY ABROAD PROGRAM | THAILAND

- Completed an intensive cybersecurity program covering networking, malware analysis, cryptography, Wi-Fi security, firewalls, honeypots, and security design principles.
- Performed hands-on labs in OSINT, reconnaissance, network traffic analysis, network scanning, password auditing (Hashcat, John the Ripper), hashing, data integrity, and steganography.

PROJECTS

Confidence-Weighted Zero-Trust Access Control | Python | Bayesian Inference | (Mar 2026 – July 2026)

- Designed a Zero Trust model computing dynamic user trust scores via Bayesian techniques, eliminating static permissions; maps to NIST SP 800-207. Weights contextual signals , device posture, behaviour, location, to evaluate risk in real time; reduces over-privileged access by design. **Published under DOI**: 10.1109/CIACON70148.2026.11688234

Ethereum Blockchain Fraud Wallet Detection | (September 2026 – Present)

- Developed a hybrid fraud-detection pipeline using Random Forest, Isolation Forest, rule-based risk scoring, anomaly detection, feature selection, and statistical analysis. Engineered wallet-level transaction features and optimized the feature set for leakage-free, interpretable fraud classification.

Noise-Aware Adaptive n-Qubit Quantum Teleportation | Reinforcement Learning | (Jan 2026 – Present)

- Developing an adaptive n-qubit quantum teleportation framework using Reinforcement Learning (PPO/DQN/SAC) over dynamic quantum channels. Incorporates channel-state estimation, circuit selection, error mitigation, entanglement optimization, and adaptive feedback using fidelity and resource-cost evaluation.

LEADERSHIP AND AWARDS

Chair, IEEE Computer Society Student Branch Chapter | IEM Kolkata | (Feb 2025 – Present)

- Organized 15 technical events for 100+ members; managed speaker sessions and cross-branch collaboration.

Awards: 2nd rank in dept. | 2nd place poster (Post-Quantum Security Threats) | 3rd place Packet Hijackers ethical hacking competition

TECHNICAL SKILLS

Subject of Interest: Data Structure (DSA), Database Management System, Cybersecurity

Programming Languages: Python, Java, C, SQL

Cybersecurity & Networking: Access Control, Network Security, Cryptography, OSINT, TCP/IP, Nmap, Wireshark

Blockchain: Ethereum, Blockchain Analytics, Wallet Analysis

Security Tools: Hashcat, John the Ripper, Steghide, AWStats

Cloud, DevOps, Web & Databases: AWS EC2, GitHub, Render, Docker, Flask, MariaDB, Supabase

AI / ML: Random Forest, Isolation Forest, Neural Networks, Bayesian Models

OS: Kali Linux, Red Hat Enterprise Linux (RHEL), Windows

Quantum Computing: Quantum Teleportation, Quantum Channels, Reinforcement Learning

CERTIFICATIONS

- Cryptography and Network Security - Indian Institute of Technology Kharagpur
- Computer Networks And Internet Protocol - Indian Institute of Technology Kharagpur
- Advanced System Security & Secure Design - University of Colorado
- Ethical Hacking & Penetration Testing - SkillUp